香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 2: Overview of control system

Yajing Shen (申亞京)

Office: 2450

eeyajing@ust.hk

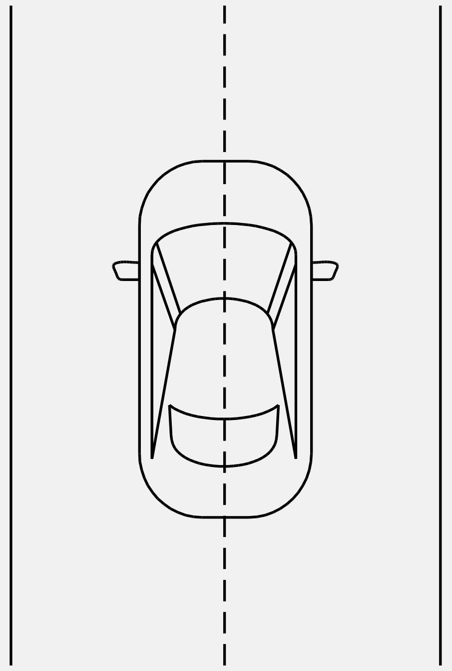
Overview

——Basic concepts

Stabilization

An example:
Driving a car along a straight road

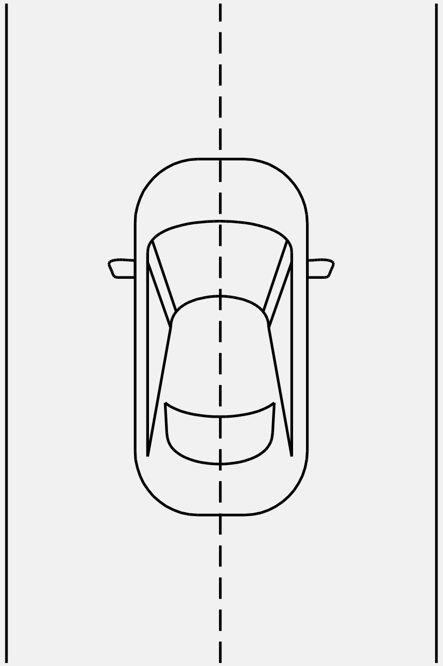
- Without a driver,
  the car will run off the road – **unstable**
- Task: **keeping the car as close to the center line as possible** – **stabilization**
- Plant: **the car**
- Controller: **the driver**



Stabilization

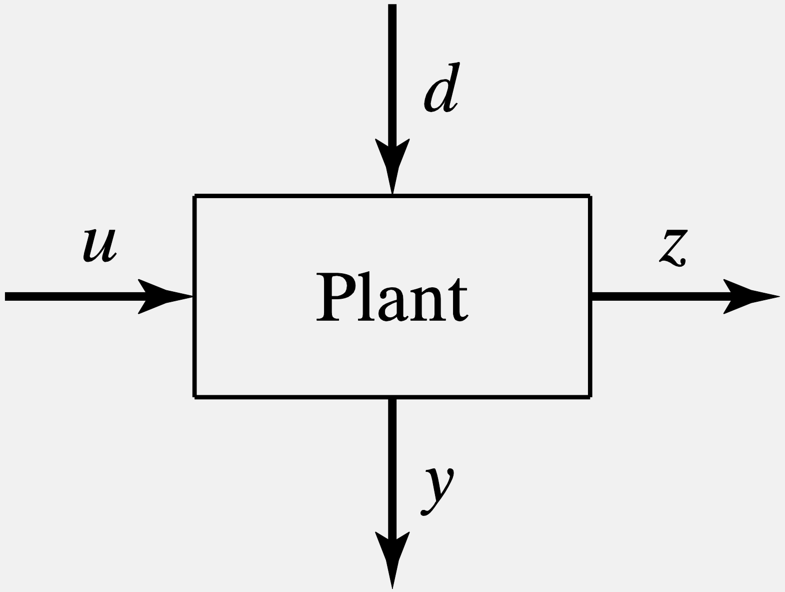
An example:
Driving a car along a straight road

- Open-loop control: **driving the car with eyes closed**
  - impossible to keep the car in the center of the road in face of uncertainty
  - impossible to stabilize an unstable plant
- Closed-loop control: **driving in the usual way**
  - turning the wheel based on observations
  - using the output information to adjust input



Stabilization

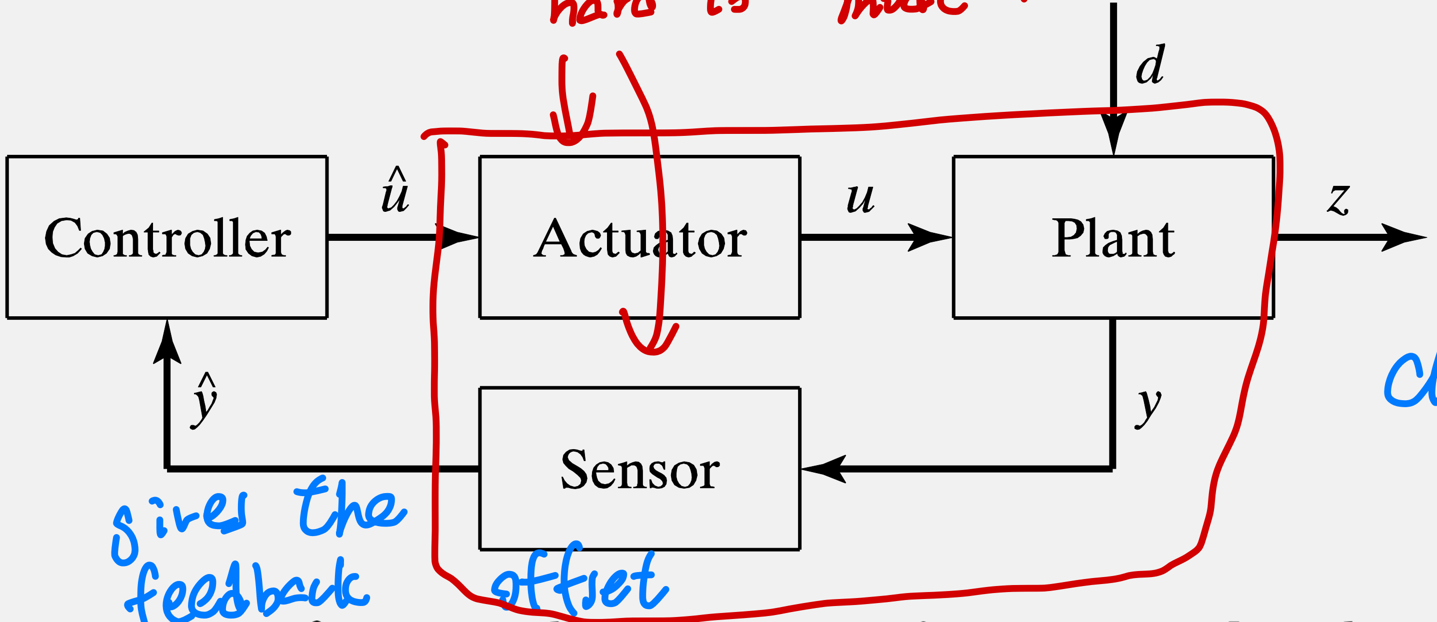
- **A block diagram for a plant**



- **Input** u: the wheel angle
- **Output** z: the deviation of the car from the center line
- **Disturbance** d: wind gust, road conditions, etc.
- **Measurement** y: available information from the car

Stabilization

hard to model!



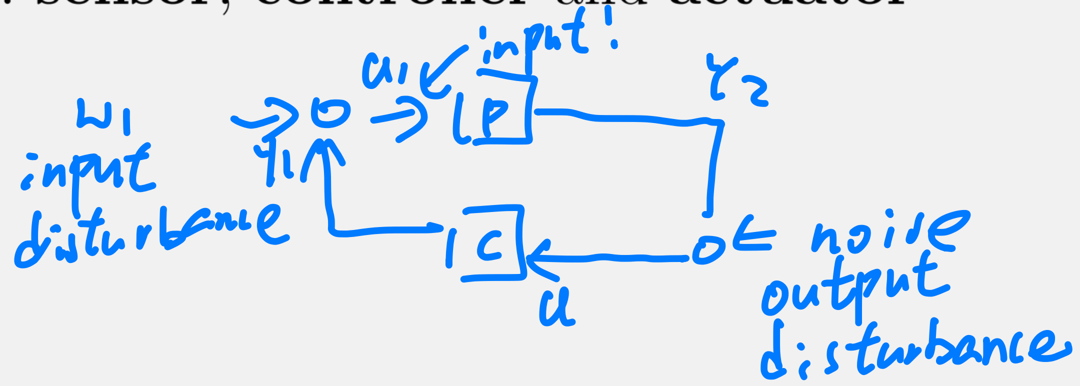
Too many things!

close loop!

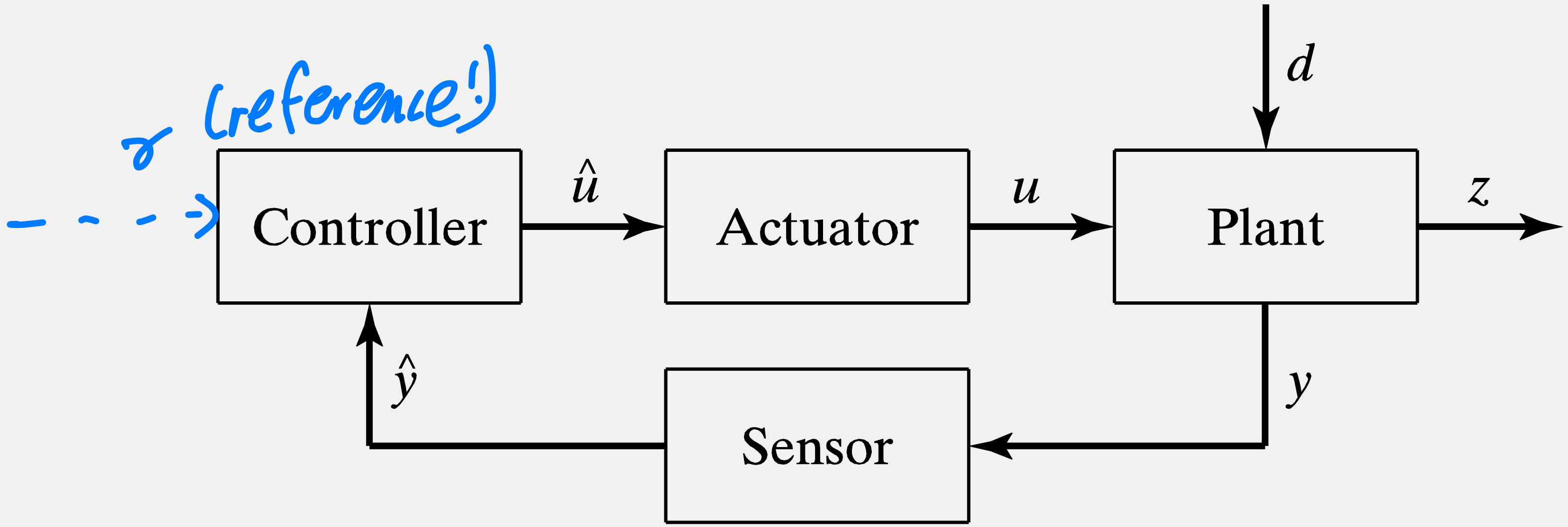
gives the feedback

offset

- Functions of a control system: **sensing**, **control** and **actuation**
- Components of a control system: **sensor**, **controller** and **actuator**



Stabilization



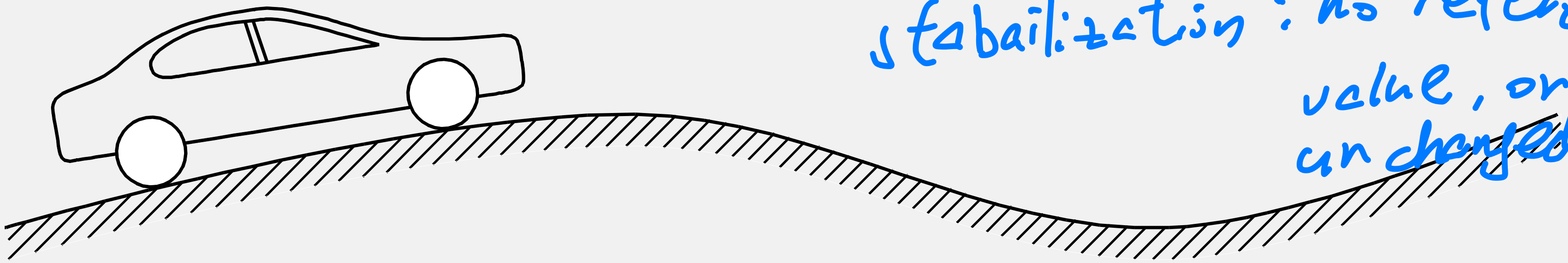
- Functions of a control system: **sensing**, **control** and **actuation**
- Components of a control system: **sensor**, **controller** and **actuator**

▶ **Stabilization problem:** Given a plant, select a sensor and an actuator, and **design a controller** so that the closed-loop system has good stability.

Regulation

Difference?

- Goal is changing

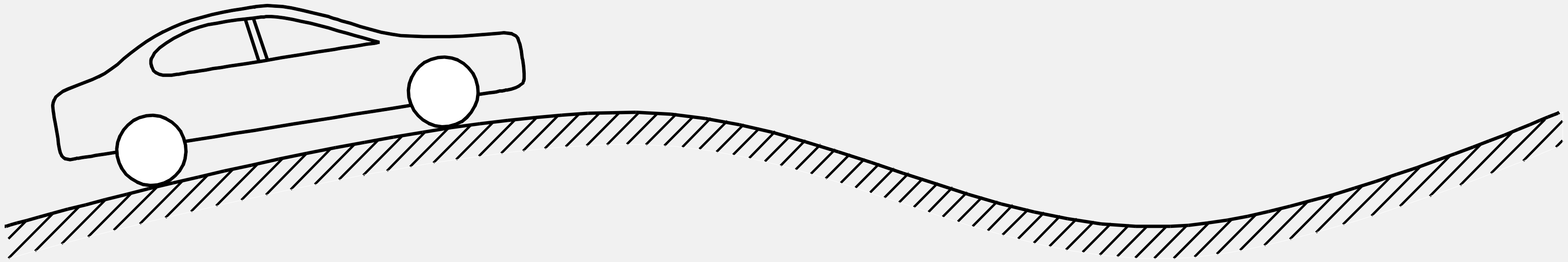


stabailization: no reference value, or unchanged

An example: Speed control of a moving car

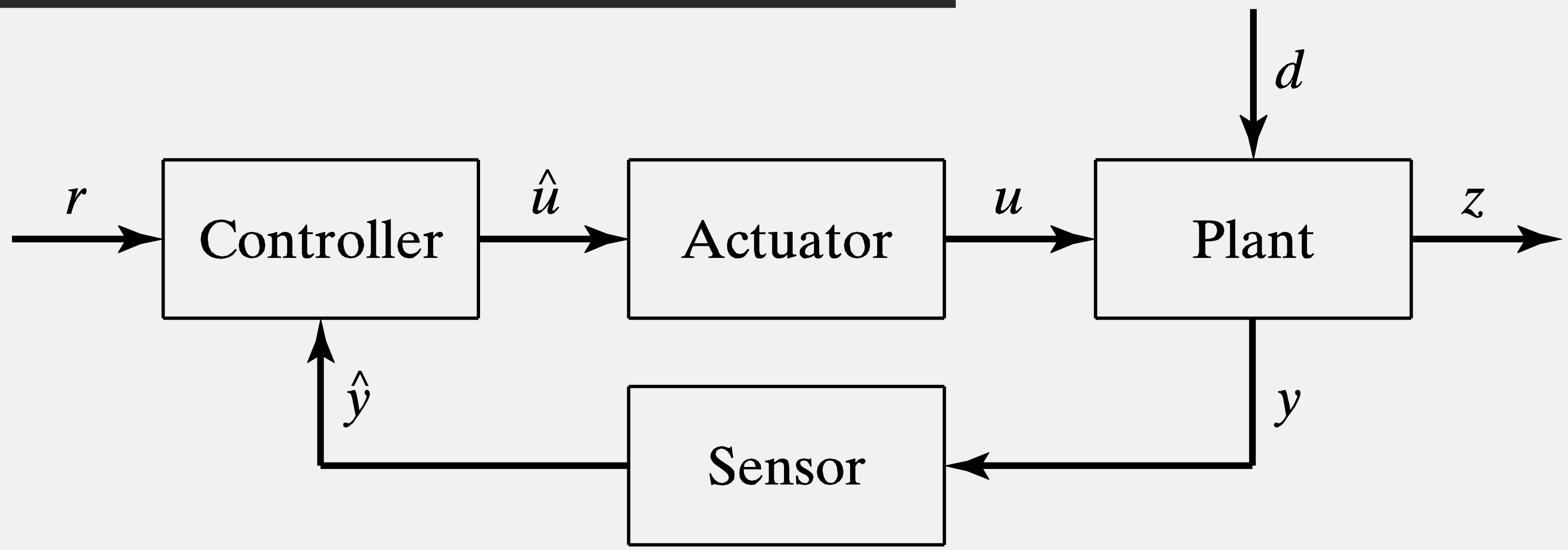
- **Tracking:** making the car speed follow an **external command**
  - set-point tracking (a piecewise constant speed command)
- **Disturbance rejection:** reducing or eliminating the effect of the **external disturbances**

Regulation



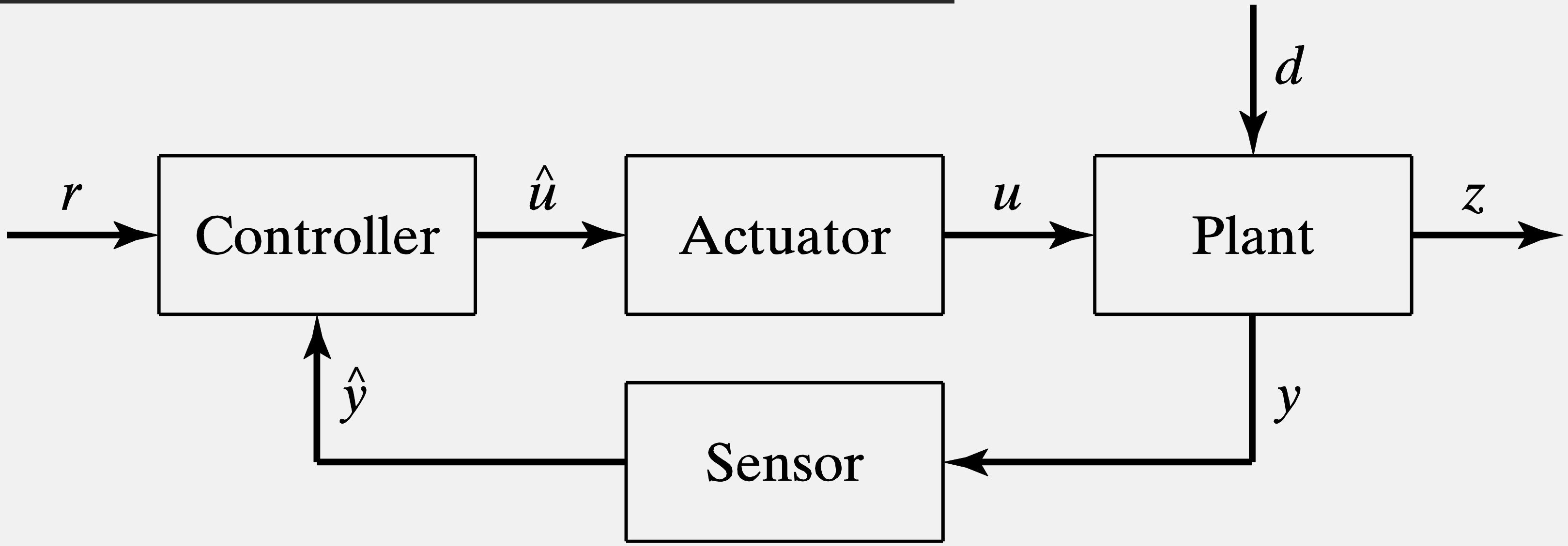
- **Open-loop control:** **setting** the gas pedal movement **in advance** according to the speed command and all the information available
  - sensitive to **uncertainties**
- **Closed-loop control:** **adjusting** the gas pedal movement according to the actual speed of the car
  - able to control the car speed so that it has a **small error**.

Regulation



- **Reference signal** r: speed command
- **Input** u: gas pedal movement
- **Output** z: acutal speed of the car
- **Disturbance** d: weather, road conditions, etc.
- **Measurement** y: available measurement

Regulation



- **Regulation problem:** Given a plant, select a sensor and an actuator, and **design a controller** so that the output of the closed-loop system could **track the reference signal**, and **eliminate the effect of disturbances**.

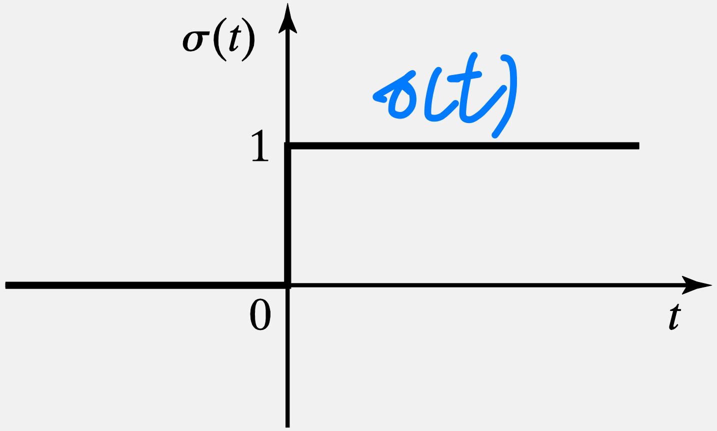
Typical Unilateral Signals

A **signal** $x(t)$ is a real-valued function of time, i.e., $x(t)\colon \mathbb{R} \to \mathbb{R}$.

Unilateral or **one-sided** signals, i.e., signals $x(t)$ with $x(t) = 0$ for all $t < 0$.

- **Unit step** signal $\sigma(t)$

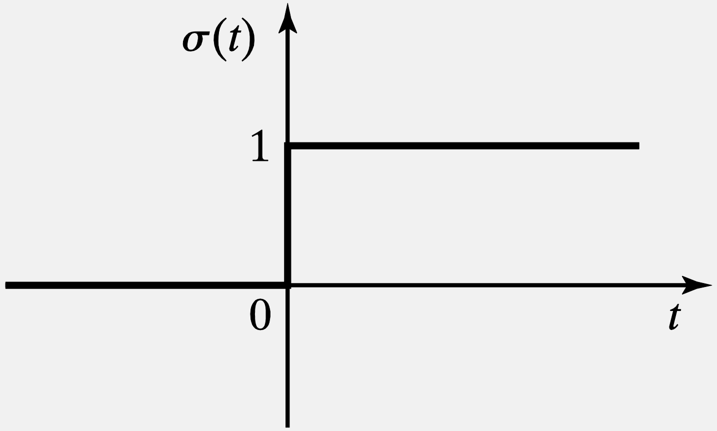
$$\sigma(t) = \begin{cases} 0 & t < 0, \\ 1 & t \geq 0. \end{cases}$$



Typical Unilateral Signals

- **Unit step** signal $\sigma(t)$

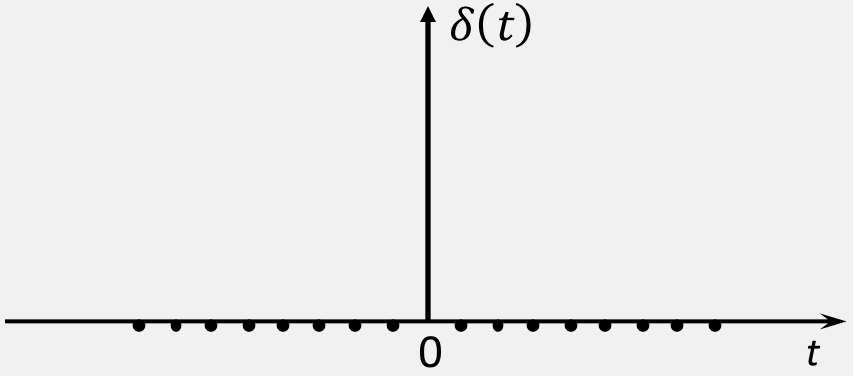
$$\sigma(t) = \begin{cases} 0 & t < 0, \\ 1 & t \geq 0. \end{cases}$$



- **Unit impulse** signal $\delta(t)$ (**singular function**)

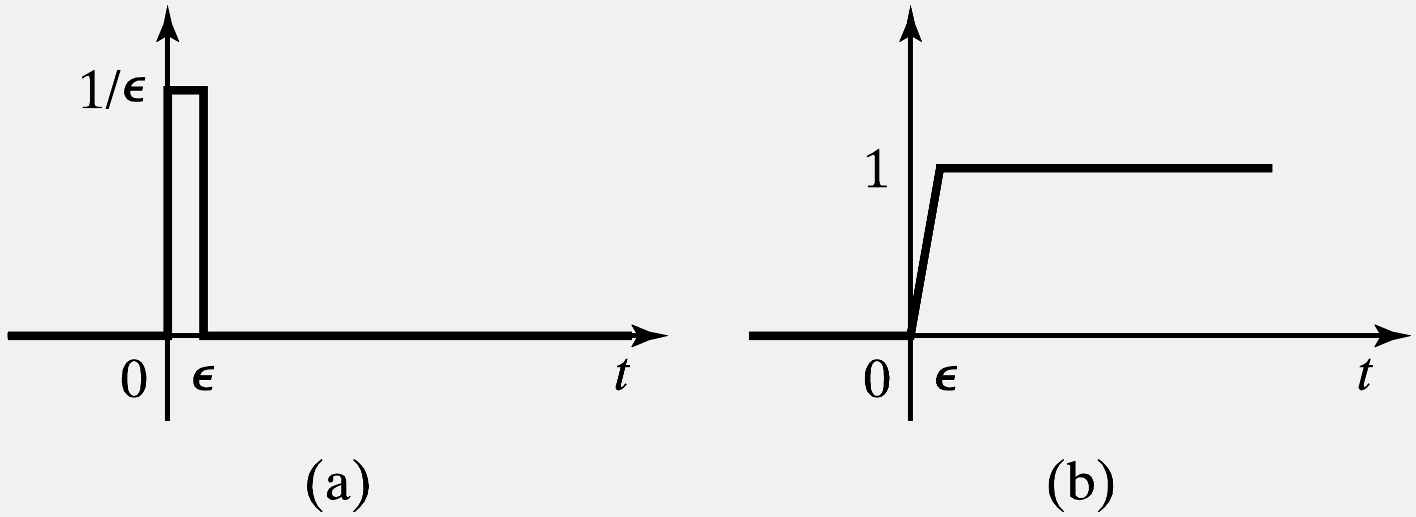
$$\begin{cases} \delta(t) = 0, & t \neq 0 \\ \int_{-\infty}^{t} \delta(\tau) d\tau = \sigma(t), & \end{cases}$$

or by the property: $\delta(t) = \dfrac{d\sigma(t)}{dt}$.



Typical Unilateral Signals

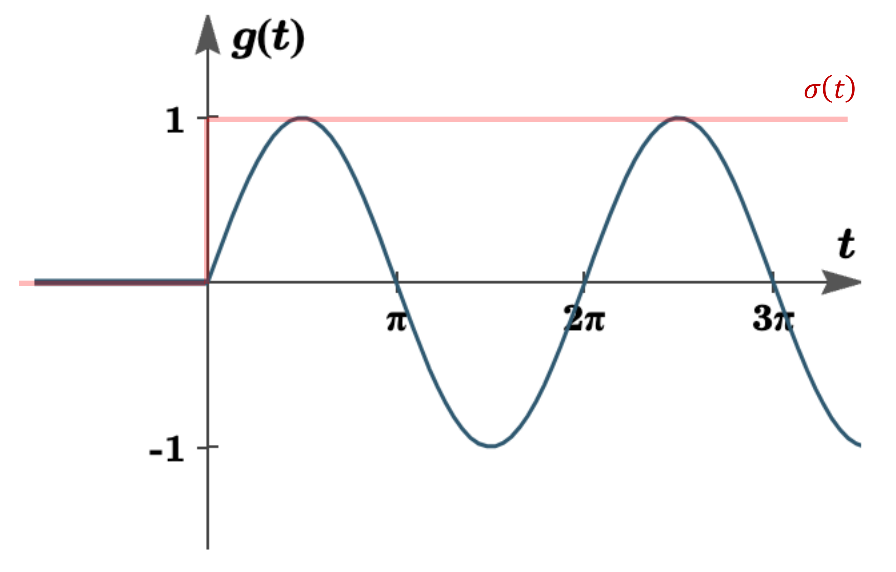
- **Approximation of** $\delta(t)$ **and** $\sigma(t)$



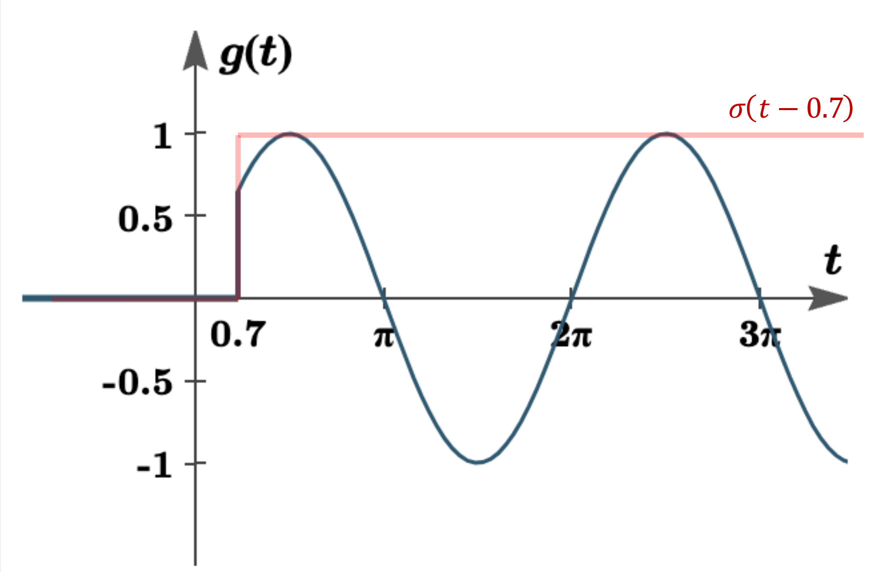
- **Other unilateral signals:** $\sin \omega t \sigma(t)$, $e^{\lambda t}\sigma(t)$
- **Other unilateral signals:** $\sin\omega t\delta(t)$, $e^{\lambda t}\delta(t)$

Typical Unilateral Signals

- **Other unilateral signals:** $\sin \omega t \sigma(t)$, $e^{\lambda t}\sigma(t)$



$sint \cdot \sigma(t)$



$sint \cdot \sigma(t - 0.7)$

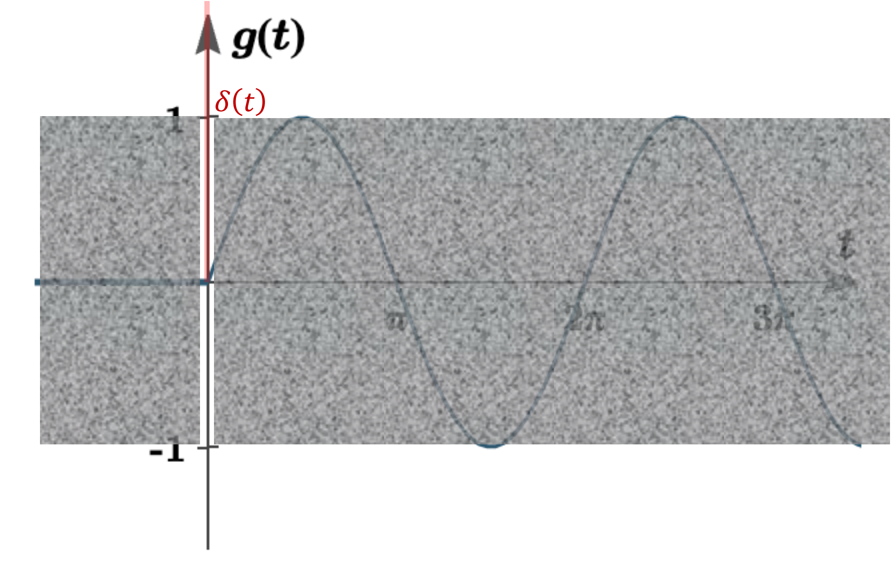
like a switch

When combine $\sigma(t)$ with other functions, the unit step function "turns off" a portion of their graph.

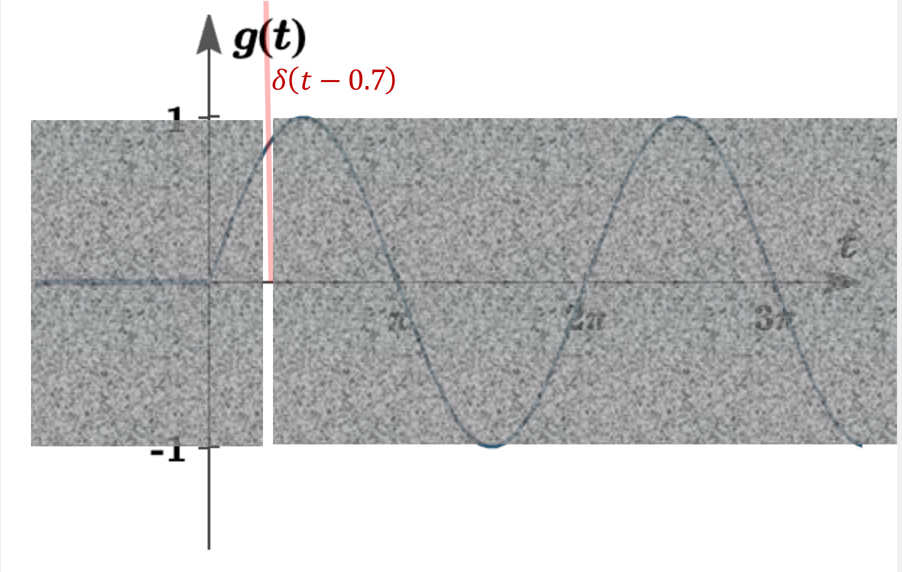
https://en.wikipedia.org/wiki/Heaviside_step_function

Typical Unilateral Signals

- **Other unilateral signals:** $\sin\omega t\delta(t)$, $e^{\lambda t}\delta(t)$



$\sin t \cdot \delta(t)$



$\sin t \cdot \delta(t-0.7)$

$$\int_{-\infty}^{\infty} f(t)\,\delta(t-T)\,dt = f(T)$$

When combine $\delta(t)$ with other functions, the unit impulse function "sifting (like a window)" their graph.

https://en.wikipedia.org/wiki/Dirac_delta_function

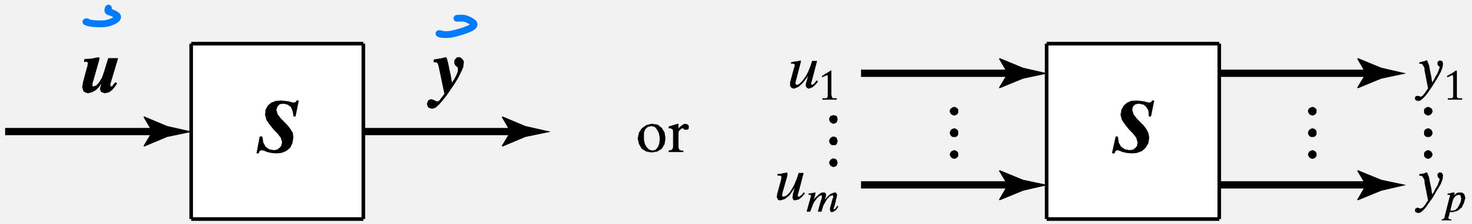
Graphic Representation of Systems

- A **system** $\boldsymbol{S}$ is a map that transforms the input $\boldsymbol{u}(t)$ to the output $\boldsymbol{y}(t)$.

$Y = SU$

$$\boldsymbol{u}(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \qquad \boldsymbol{y}(t) = \boldsymbol{S}[\boldsymbol{u}(t)] \qquad \boldsymbol{y}(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix}$$

$$\boldsymbol{y} = \boldsymbol{S}\boldsymbol{u}$$



song you hear
temperature of engine
speed
direction
⋮

$U = [u_1, u_2, u_3 \cdots u_m]$
$Y = [Y_1, Y_2 \cdots Y_n]$
no need same dimension vectors

System Classifications

- **According to numbers of input and output channels**
  - **Single-input-single-output (SISO) system**
  - Multi-input-single-output (MISO) system
  - Single-input-multi-output (SIMO) system
  - Multi-input-multi-output (MIMO) system
- For technical reasons, we will mostly study SISO control systems.
- Only occasionally will we deal with MISO and SIMO systems.

System Classifications

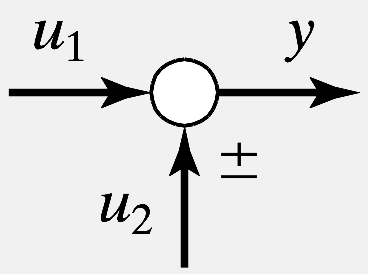
- **According to memory**
  - Static systems (memoryless): **P**roportional systems, i.e., $y(t) = K_P \boxed{u(t)}$ Present
  - **Dynamic systems:** **I**ntegral systems, i.e., $y(t) = K_I \int_{-\infty}^{t} u(\tau)d\tau$ Past
- **According to causality**
  - **Causal systems:** **P**roportional-**I**ntegral systems, i.e.,

$$y(t) = K_P u(t) + K_I \int_{-\infty}^{t} u(\tau)d\tau$$

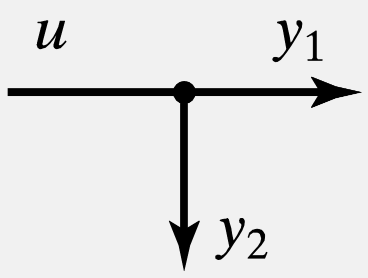
  - Noncausal systems: **D**erivative systems, i.e., $y(t) = K_D \frac{du(t)}{dt}$ Decision from Future

System Classifications

- **Two special MISO and SIMO systems**



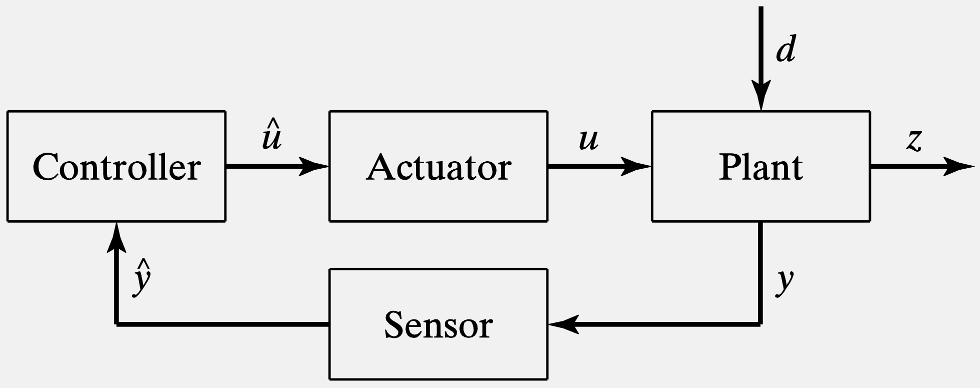
Summing point: $y = u_1 \pm u_2$



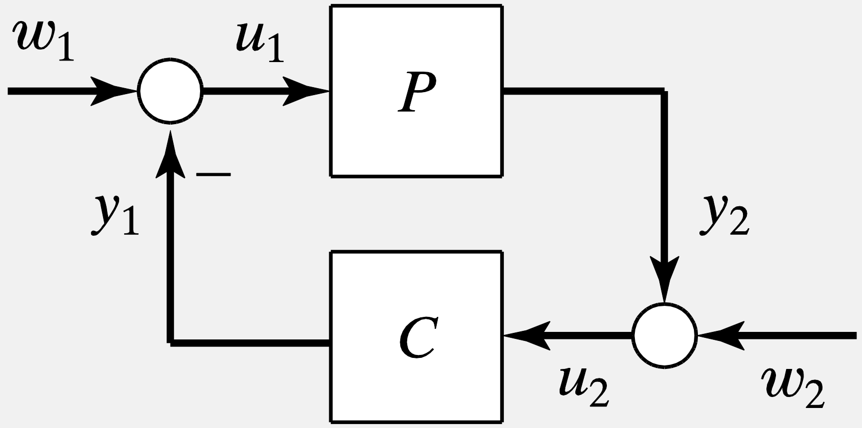
Pickoff point: $y_1 = u, y_2 = u$

Basic Structures

- **Feedback system for stabilization**



A simpler yet more abstract structure



This structure is not very convenient for a theory:

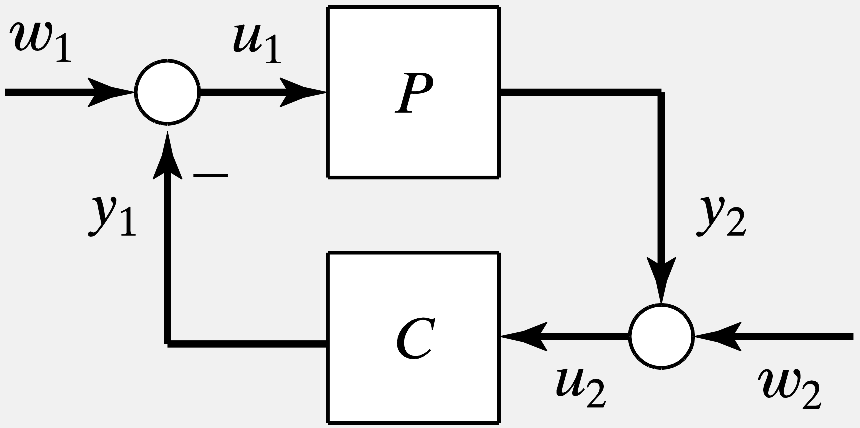
- There is no theory for the selection of actuators and sensors.
- The effect of disturbance is very difficult to model.
- In most of the applications, taking the measurement the same as the output has important advantages.

Therefore, we:

- Absorb the sensors and actuators into the plant
- Simplify the way the disturbance enters the system so that there is only an input disturbance and an output disturbance
- Assume that the measurement and the output are the same.

Basic Structures

- **Feedback system for stabilization**

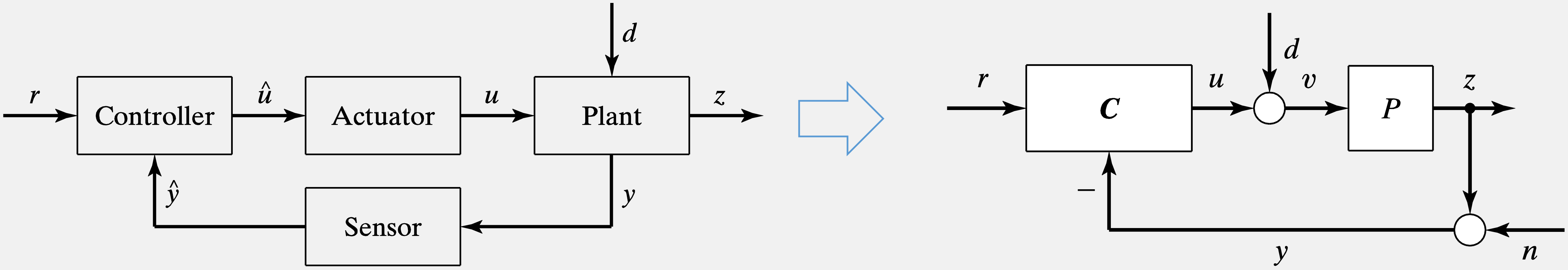


Stabilization problem: Given plant P, design controller C so that the system has **good stability**.

Basic Structures

- **Feedback system for regulation**

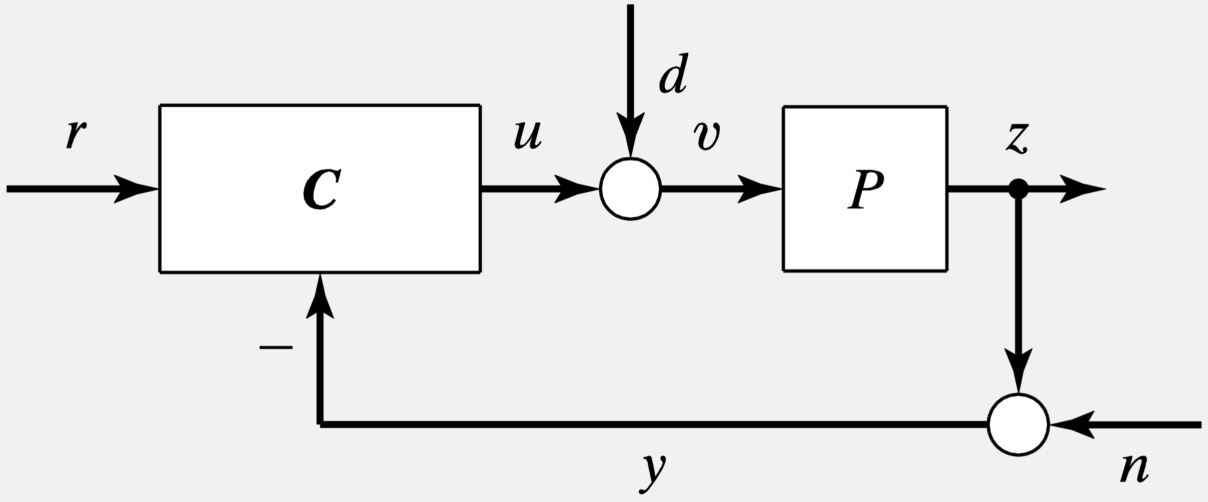
A simpler yet more abstract structure



Similarly, we also absorb the sensor and actuator to the plant and group the disturbances and noises into two groups: input disturbances and output noises.

Basic Structures

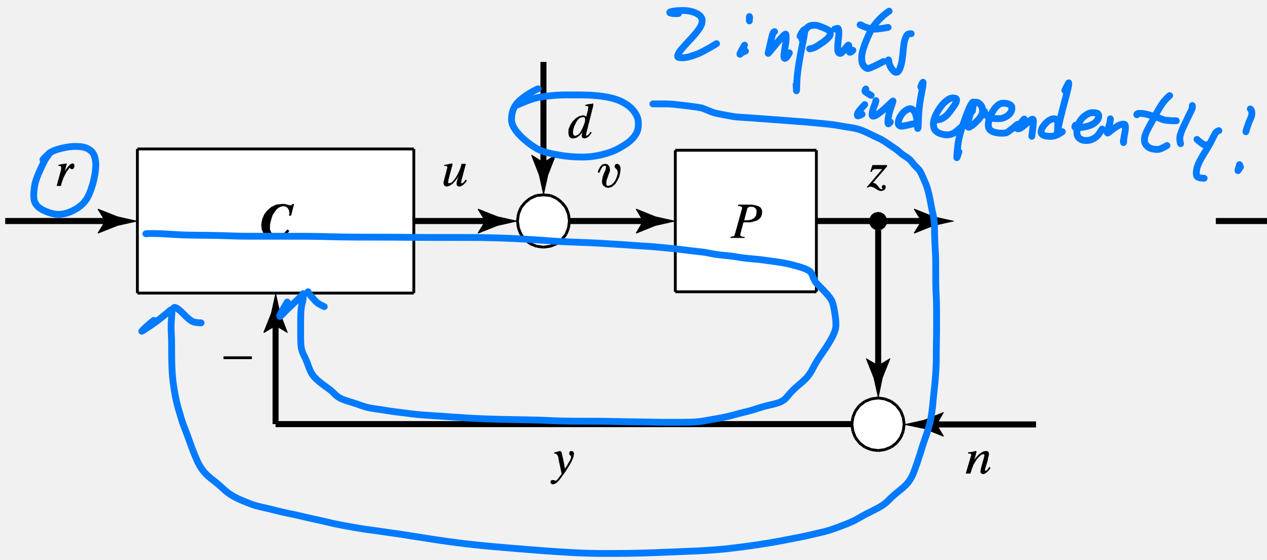
- **Feedback system for regulation**

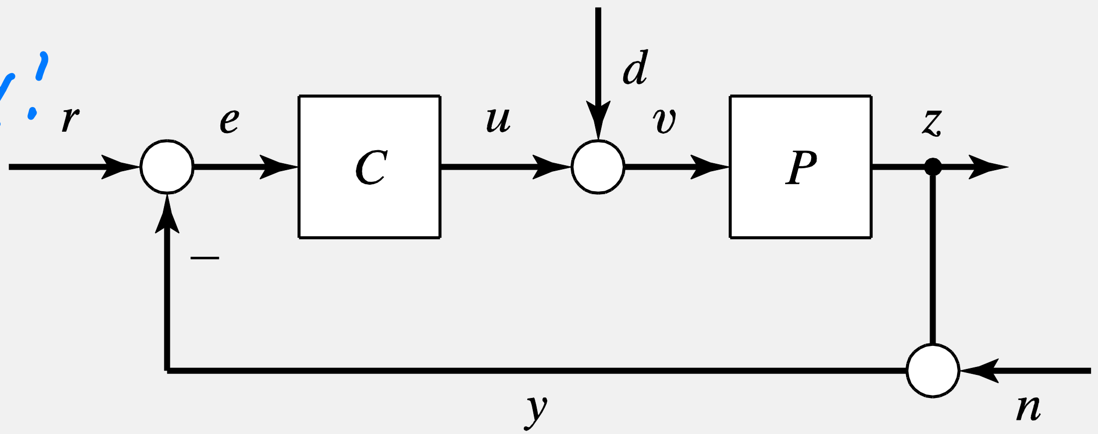


Regulation problem: Given plant P, design two-degree-of-freedom (2DOF) controller $\boldsymbol{C} = \begin{bmatrix} C_1 & C_2 \end{bmatrix}$ so that **good performance** in **tracking** and **disturbance rejection** is achieved.

2DOF Controller

in L15!





- 2DOF controller provides significant advantages in
  - achieving better performance
  - **trading off** the often conflicting **tracking** and **disturbance rejection** requirements

- 1DOF structure is actually a special case of the 2DOF structure by setting

$$\boldsymbol{C}\begin{bmatrix} r \\ -y \end{bmatrix} = C(r-y)$$

Case Study: A MagLev System

Case Study: A MagLev System

input: voltage

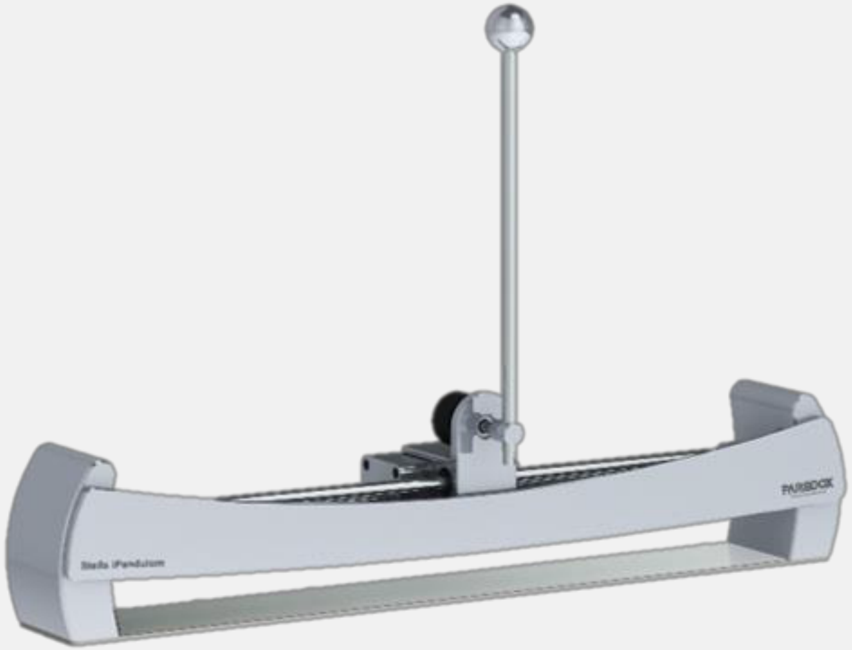
output: position

- **Task:** suspend the steel ball at **a desired position**.
- **Questions:** SISO, SIMO, MISO or MIMO?
  Stabilization or regulation?

Ans:
SISO: Input: voltage, output: position.
Stabilization: keep it at a desired position.

Case Study: An iPendulum System

Case Study: An iPendulum System



movement speed

- **Task:** keep the pendulum **upward** by moving the car at the base.
- **Questions:** SISO, SIMO, MISO or MIMO? Stabilization or regulation?

Ans:
SIMO: Input: force applied on the car, output: position and angle of the pendulum.
Stabilization: keep it upward.

Review & Exercise

- Can you tell the difference of “Stabilization” and “Regulation”?
- What’s the characteristic of unit step signal and unit impulse signal?

u Y S

- What is input, output and system? $Y = Su$
- How to represent a control system through equation and graph?

I

- How to classify a system according to memory and causality?
- What does SISO mean in control system?
- Can you explain the SISO, MISO, SIMO and MIMO system with examples?
- How to represent the feedback system for stabilization and regulation?